

- 6 (a) unit of magnetic flux density / magnetic field strength B1
 (uniform) field normal to wire carrying current of 1 A M1
 giving force (per unit length) of 1 N m^{-1} A1 [3]
- (b) (i) force on magnet / balance is downwards (so by Newton's third law) B1
 force on wire is upwards M1
 pole P is a north pole A1 [3]
- (ii) $F = BIL$ and $F = mg$ (g missing, then 0/3 in (ii)) C1
 $2.3 \times 10^{-3} \times 9.8 = B \times 2.6 \times 4.4 \times 10^{-2}$ ($g = 10$, loses this mark) C1
 $B = 0.20 \text{ T}$ A1 [3]
- (c) reading for maximum current = $2.3 \times \sqrt{2}$ C1
 total variation = $2 \times 2.3 \times \sqrt{2}$
 = 6.5 g A1 [2]
- 7 coil in series with meter (*do not allow inclusion of a cell*) B1